

OASIS Agentic Pipeline

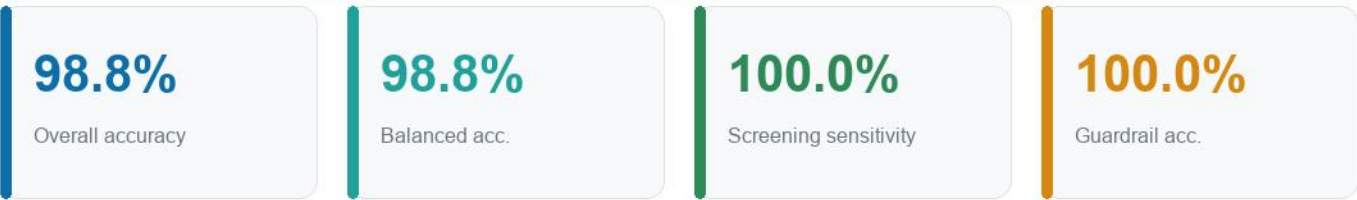
Effectiveness & Data Analysis Report

Automated evaluation report generated 2026-06-17.

- MRI scans evaluated: 160 (stratified across 4 classes)
- Clinical cohort: 100 subjects | Longitudinal records: 373
- Acceleration: ONNX Runtime QNN (Snapdragon NPU) → DirectML → CPU fallback
- Language reasoning: local Ollama (no paid API tokens)

Hybrid edge-cloud, Snapdragon-NPU-optimized, zero paid-API multi-agent system for Alzheimer's disease screening on the OASIS datasets.

1. Executive Summary



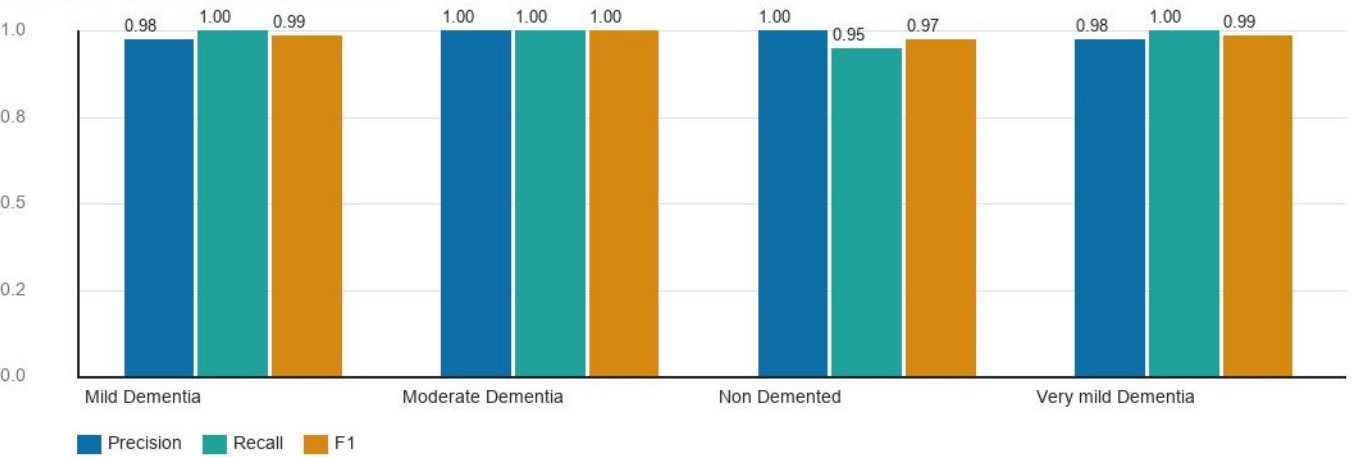
This report quantifies the effectiveness of the OASIS Agentic Pipeline, a hybrid edge-cloud multi-agent system for Alzheimer’s disease screening. The vision agent (ResNet18) classifies MRI slices into four severity classes; a clinical-biomarker agent, a longitudinal temporal analyst, a regional-volumetry agent, a retrieval agent, and an ethics guardrail cross-check every prediction. A local Ollama-backed reasoner produces the final grounded narrative — entirely on free, on-device models.

- Cohen's kappa (chance-corrected agreement): 0.983.
- Mean model confidence on evaluated slices: 92.5%.
- Ethics guardrail correctly adjudicated 8/8 adversarial safety cases.

2. Vision Agent Classification Performance

Evaluated on 160 held-out MRI slices using the training-time validation transform (grayscale, 224×224, normalized). Class indices follow the alphabetical ImageFolder ordering used during training.

Per-class precision / recall / F1



Class	Precision	Recall	F1	Support
Mild Dementia	0.98	1.00	0.99	40
Moderate Dementia	1.00	1.00	1.00	40
Non Demented	1.00	0.95	0.97	40
Very mild Dementia	0.98	1.00	0.99	40

2.1 Confusion Matrix

Confusion matrix (true vs predicted)

Mild Dementia	40100%	00%	00%	00%
Moderate Dementia	00%	40100%	00%	00%
Non Demented	12%	00%	3895%	12%
Very mild Dementia	00%	00%	00%	40100%
	Mild Dementia	Moderate Dementia	Non Demented	Very mild Dementia

Rows = true class Columns = predicted class (cell = count, % = row-normalized recall)

2.2 Clinical screening view

Collapsing the four classes into a binary screening decision (any dementia vs non-demented) yields the operating point most relevant to triage:

- Sensitivity (impaired correctly flagged): 100.0%
- Specificity (healthy correctly cleared): 95.0%

3. Cohort & Longitudinal Data Analysis

Cross-sectional cohort: 100 subjects. Age 74.8 ± 7.3 yr; MMSE 24.9 ± 3.3 ; mean nWBV 0.729.

- Sex distribution: F=56, M=44

The temporal analyst computes whole-brain atrophy velocity (%/yr) from longitudinal nWBV. Group separation validates the longitudinal agent:

Mean atrophy velocity by diagnostic group (%/yr)



4. Ethics Guardrail Effectiveness

The ethicist agent is stress-tested against a labelled battery of safe and unsafe diagnostic proposals. It must flag unsafe/contradictory cases and pass consistent ones.

100%

Battery accuracy

8

Cases tested

Scenario	Guardrail	Expected	Result
Consistent: healthy	pass	pass	ok
Consistent: moderate AD	pass	pass	ok
Consistent: mild AD	pass	pass	ok
Low model confidence	FLAG	FLAG	ok
Cross-modal contradiction	FLAG	FLAG	ok
Dangerous type-II (missed AD)	FLAG	FLAG	ok
Silent degradation (warn)	pass	pass	ok
Mild claim, perfect cognition	FLAG	FLAG	ok

5. Regional Volumetry (FreeSurfer aseg)

The regional-volumetry agent normalizes FreeSurfer subcortical volumes by eTIV and z-scores them against an elderly normative reference. Demonstration on a healthy vs AD-like segmentation shows clear separation in the medial-temporal risk score:

Subject	Hippocampus z	Ventricle z	MTA risk	Stage
Healthy reference	+0.05	+0.09	0.04	Within normal limits
Atrophic (AD-like)	-2.24	-2.09	2.12	Severe medial-temporal atrophy (MTA-like)

Medial-temporal-atrophy risk score



6. Edge Acceleration Benchmark (Snapdragon NPU)

96.3 MB

FP32 model

12.1 MB

INT8 model

87%

Size reduction

Per-inference latency by execution provider (ms, lower is better)



Execution provider	Latency	Status
CPU	28.2 ms	available
PyTorch FP32 (reference)	26.2 ms	baseline

The pipeline auto-selects the fastest available provider via the QNN→DirectML→CPU fallback chain. On Snapdragon X hardware the QNN row maps to the Hexagon NPU; on other machines it gracefully degrades to DirectML or CPU.

7. Methodology & Limitations

- Vision metrics: stratified, class-balanced HELD-OUT sample (seed=42, disjoint from the balanced-trainer's per-class training window when max-per-class ≤ 80), training-consistent normalization, evaluated on CPU.
- Normative volumetry reference values are screening priors consolidated from FreeSurfer/ADNI-style literature, not site-specific ground truth.
- Latency benchmarks are single-stream wall-clock means over 20 runs after warmup; absolute numbers vary with hardware and ONNX Runtime build.
- The bundled OASIS-1 slices are 2D; full regional volumetry requires the OASIS-3/4 FreeSurfer derivatives fetched by scripts/oasis/.
- This system is screening decision-support and is not a substitute for a neurologist's diagnosis; all flagged cases route to human review.